

Chapter 8

Convex Geometry

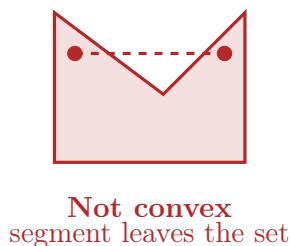
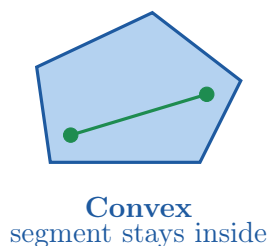
In many practical problems, there is more than one variable to consider and more than one condition that must be satisfied. For example, a business may produce two products while facing limits on labor, materials, storage, or budget. Each condition creates a constraint, and the solutions that satisfy all of the constraints together form a region in the plane.

This chapter introduces the geometry of such regions. We study half-planes, feasible regions, and convex sets, with special attention to the polygons that arise from systems of linear inequalities. These ideas provide the geometric foundation for linear programming in the next chapter.

8.1 Convex Sets

Definition. *Convex set.* A set of points in the plane is **convex** if the line segment connecting any two points in the set lies entirely within the set.

Convexity is a property of the entire set. In practice, the examples that appear most often in this course are convex polygons, and they can usually be recognized visually by the fact that the boundary does not bend inward.



Standard convex shapes include line segments, triangles, rectangles, regular polygons, circles, and half-planes. A half-plane is the set of all points on one side of a line, including the line itself.

Example 8.1.1. Determine whether each set is convex.

- (a) A solid triangle. *Convex*. Any line segment connecting two points inside a triangle stays inside it.
- (b) An L-shaped region. *Not convex*. A segment connecting the two outer corners of the L passes through the missing interior section.
- (c) The set of all points satisfying $x \geq 0$ and $y \geq 0$ (the first quadrant, including the axes). *Convex*. Any segment connecting two first-quadrant points stays in the first quadrant.
- (d) A donut shape (ring). *Not convex*. A segment connecting two points on opposite sides of the ring passes through the hole.

Exercises 8.1

8.1.1. Determine whether each shape is convex or not. For each non-convex shape, describe a specific line segment that leaves the set.

- (a) A solid rectangle.
- (b) A crescent (moon-shaped region).
- (c) A solid equilateral triangle.
- (d) A figure-eight shape.
- (e) The set of all points with $x \geq 2$.
- (f) A star with five pointed arms.

8.1.2. Explain in one complete sentence why a line segment (considered as a filled shape) is convex.

8.1.3. The intersection of two convex sets is always convex. Verify this claim for the intersection of the half-planes $x \geq 0$ and $y \geq 0$. Sketch the intersection and confirm that any segment between two points in it stays inside.

8.1.4. A student claims that the union of two convex sets is always convex. Provide a counterexample using two simple shapes to show this claim is false.

8.2 Linear Inequalities and Half-Planes

A **linear inequality** in two variables x and y is an expression of the form:

$$ax + by \leq c, \quad ax + by \geq c, \quad ax + by < c, \quad \text{or} \quad ax + by > c.$$

Each inequality divides the plane into two regions: the **boundary line** $ax + by = c$, and the two **half-planes** on either side of it.

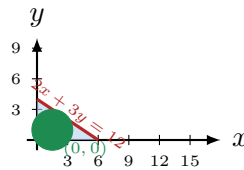
To determine which half-plane satisfies the inequality, substitute a test point. The point $(0, 0)$ is convenient when it does not lie on the boundary line.

Example 8.2.1. Sketch the half-plane satisfying $2x + 3y \leq 12$.

Step 1: Draw the boundary line $2x + 3y = 12$. Find two points: at $x = 0$, $y = 4$; at $y = 0$, $x = 6$. Draw the line through $(0, 4)$ and $(6, 0)$.

Step 2: Test the point $(0, 0)$. $2(0) + 3(0) = 0 \leq 12$. ✓

The origin satisfies the inequality, so shade the half-plane containing the origin.



Example 8.2.2. Sketch the half-plane satisfying $x + y \geq 5$.

Boundary line: $x + y = 5$, passing through $(0, 5)$ and $(5, 0)$.

Test $(0, 0)$: $0 + 0 = 0 \not\geq 5$. The origin does not satisfy the inequality, so shade the half-plane not containing the origin: the region above and to the right of the line.

Example 8.2.3. (*Non-negativity constraints*) The constraints $x \geq 0$ and $y \geq 0$ restrict the region to the first quadrant. These are called **non-negativity constraints** and appear in nearly every applied problem in this course, because quantities such as items produced or hours worked cannot be negative.

Exercises 8.2

8.2.1. For each inequality, identify two points on the boundary line and determine which side contains the origin.

- (a) $x + 2y \leq 8$
- (b) $3x + y \geq 9$
- (c) $2x + 5y \leq 20$
- (d) $x + y \geq 6$

8.2.2. Sketch the half-plane for each inequality. Label the boundary line and shade the feasible side.

- (a) $x + y \leq 10$
- (b) $2x + y \leq 12$

(c) $x + 3y \geq 15$

8.2.3. A manufacturing constraint requires that the number of hours on Machine A (x) and Machine B (y) satisfy $3x + 2y \leq 60$.

- (a) Find the intercepts of the boundary line.
- (b) Does the point $(10, 10)$ satisfy the constraint? Show your calculation.
- (c) Does the point $(15, 8)$ satisfy the constraint?

8.2.4. The constraint $x + y \leq 8$ combined with $x \geq 0$ and $y \geq 0$ creates a triangular region.

- (a) Sketch this region.
- (b) Identify its three vertices.
- (c) Is the point $(3, 4)$ inside the region?

8.3 Feasible Regions

Definition. *Feasible region.* The **feasible region** of a system of linear inequalities is the set of all points that satisfy every inequality simultaneously.

The feasible region is the intersection of a collection of half-planes. The intersection of convex sets is convex, so the feasible region is always convex. When the feasible region is bounded (it does not extend to infinity), it is a convex polygon whose sides lie on the boundary lines of the constraints.

The **vertices** of the feasible region are the corner points where pairs of boundary lines intersect. They are found by solving each pair of boundary equations as a system of two equations in two unknowns.

Example 8.3.1. Find the feasible region and its vertices for:

$$3x + 2y \leq 120, \quad x + 2y \leq 60, \quad x \geq 0, \quad y \geq 0.$$

Step 1: Find the boundary line intercepts.

- $3x + 2y = 120$: passes through $(40, 0)$ and $(0, 60)$.
- $x + 2y = 60$: passes through $(60, 0)$ and $(0, 30)$.

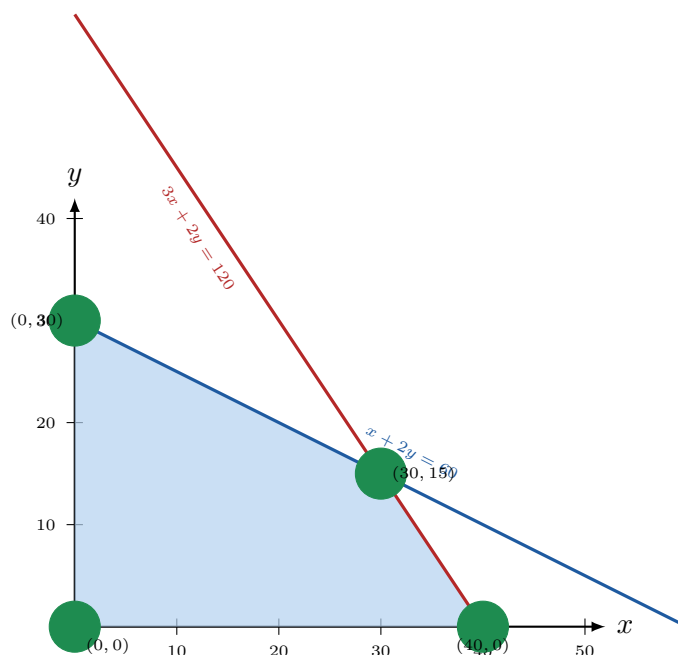
Step 2: Find the intersection of the two constraint lines. From $x + 2y = 60$: $x = 60 - 2y$. Substitute into $3x + 2y = 120$:

$$3(60 - 2y) + 2y = 120 \Rightarrow 180 - 6y + 2y = 120 \Rightarrow -4y = -60 \Rightarrow y = 15.$$

Then $x = 60 - 2(15) = 30$. Intersection: $(30, 15)$.

Step 3: Identify the four vertices.

- $(0, 0)$: origin, satisfies all constraints.
- $(40, 0)$: on the x -axis, from $3x = 120$.
- $(30, 15)$: intersection of the two constraint lines.
- $(0, 30)$: on the y -axis, from $2y = 60$.



Example 8.3.2. (*Checking a point for feasibility*) Is the point $(20, 20)$ in the feasible region from Example 8.3.1?

Check each constraint:

$$3(20) + 2(20) = 60 + 40 = 100 \leq 120 \checkmark$$

$$20 + 2(20) = 20 + 40 = 60 \leq 60 \checkmark$$

$$20 \geq 0 \checkmark$$

$$20 \geq 0 \checkmark$$

All constraints are satisfied, so $(20, 20)$ is feasible.

Is $(25, 20)$ feasible?

$$3(25) + 2(20) = 75 + 40 = 115 \leq 120 \checkmark$$

$$25 + 2(20) = 25 + 40 = 65 \not\leq 60.$$

The second constraint is violated, so $(25, 20)$ is not in the feasible region.

Exercises 8.3

8.3.1. For each system of constraints, find all vertices of the feasible region. Show the intersection calculation for any vertex that is not on an axis.

(a) $x + y \leq 8, \ x \geq 0, \ y \geq 0$

(b) $2x + y \leq 10, \ x + y \leq 7, \ x \geq 0, \ y \geq 0$

(c) $x + 3y \leq 18, \ 2x + y \leq 16, \ x \geq 0, \ y \geq 0$

8.3.2. A workshop makes chairs (x) and tables (y). Two constraints apply:

$$2x + 4y \leq 40 \quad (\text{wood available})$$

$$x + y \leq 14 \quad (\text{labor hours available})$$

with $x \geq 0$ and $y \geq 0$.

(a) Find the intersection of the two constraint lines.

(b) List all four vertices of the feasible region.

(c) Check whether $(8, 5)$ is feasible.

(d) Check whether $(10, 5)$ is feasible.

8.3.3. A point $(a, 6)$ lies on the boundary of the feasible region defined by $x + 2y \leq 20$ and $x + y \leq 15$. If this point is the intersection of the two boundary lines, find a .

8.3.4. Explain in two complete sentences why the feasible region of a system of linear inequalities is always convex. Use the definition of convexity in your explanation.

8.4 The Corner Point Principle

The Corner Point Principle. The maximum and minimum values of any linear expression $f(x, y) = ax + by$ over a bounded convex polygon are achieved at one of the vertices of the polygon.

This principle reduces the problem of finding a maximum or minimum over an infinite set of points to evaluating the expression at a finite number of vertices. The method is: find all vertices, evaluate the expression at each one, and compare.

Example 8.4.1. Maximize $\pi = 4x + 5y$ over the feasible region from Example 8.3.1 with vertices $(0, 0)$, $(40, 0)$, $(30, 15)$, $(0, 30)$.

Vertex (x, y)	Expression $\pi = 4x + 5y$	
$(0, 0)$	$4(0) + 5(0) = 0$	
$(40, 0)$	$4(40) + 5(0) = 160$	
$(30, 15)$	$4(30) + 5(15) = 195$	$\leftarrow$ maximum
$(0, 30)$	$4(0) + 5(30) = 150$	

The maximum value of $\pi = 195$ occurs at $(30, 15)$.

Example 8.4.2. Minimize $C = 2x + 3y$ over the same feasible region.

Vertex (x, y)	Expression $C = 2x + 3y$	
$(0, 0)$	$2(0) + 3(0) = 0$	$\leftarrow$ minimum
$(40, 0)$	$2(40) + 3(0) = 80$	
$(30, 15)$	$2(30) + 3(15) = 105$	
$(0, 30)$	$2(0) + 3(30) = 90$	

The minimum value of $C = 0$ occurs at the origin $(0, 0)$. This corresponds to producing nothing, which has zero cost but also zero revenue. In applied problems, additional constraints or context often rule out the origin as a practical solution.

Example 8.4.3. (*Evaluating a different objective*) Maximize $R = 7x + 9y$ over the same feasible region.

Vertex (x, y)	Expression $R = 7x + 9y$	
$(0, 0)$	$7(0) + 9(0) = 0$	
$(40, 0)$	$7(40) + 9(0) = 280$	
$(30, 15)$	$7(30) + 9(15) = 345$	
$(0, 30)$	$7(0) + 9(30) = 270$	$\leftarrow$ check

Wait: at $(30, 15)$: $R = 210 + 135 = 345$. At $(0, 30)$: $R = 270$. The maximum is $R = 345$ at $(30, 15)$.

Comparing Examples 8.4.1 and 8.4.3: both $\pi = 4x + 5y$ and $R = 7x + 9y$ achieve their maxima at the same vertex $(30, 15)$ for this particular feasible region. Different expressions, however, can achieve their maxima at different vertices, as Example 8.4.2 shows. The feasible region is fixed; the expression determines which vertex is optimal.

Exercises 8.4

8.4.1. The feasible region has vertices $(0, 0)$, $(6, 0)$, $(4, 3)$, and $(0, 5)$. Evaluate each expression at all four vertices and identify the maximum and minimum.

(a) $f = 3x + 2y$

(b) $f = x + 4y$

(c) $f = 5x + y$

8.4.2. The feasible region has vertices $(0, 0)$, $(8, 0)$, $(5, 4)$, and $(0, 6)$.

(a) Maximize $\pi = 6x + 8y$.

(b) Minimize $C = 3x + 5y$ (ignoring the origin if it corresponds to zero production).

(c) Does the expression $f = 4x + 4y$ achieve its maximum at a unique vertex? Evaluate it at all four vertices and explain your answer.

8.4.3. A bakery produces loaves (x) and muffin trays (y) subject to the constraints in Example 8.3.1. The profit is $\pi = 5x + 8y$.

(a) Evaluate π at each vertex.

(b) Identify the production plan that maximizes profit.

(c) How does the optimal vertex for $\pi = 5x + 8y$ compare to the optimal vertex for $\pi = 4x + 5y$ from Example 8.4.1?

8.4.4. Explain in two complete sentences why the Corner Point Principle simplifies the problem of maximizing a linear expression. What would be required to find the maximum without this principle?

8.5 From Chapter 7 to Chapter 8

Chapter 7 introduced the break-even analysis for a single product. The profitable zone was all quantities $q > q^*$: a half-line, which is the simplest convex set. Chapter 8 extends this to two products and multiple resource constraints.

The connection is direct. In Chapter 7, the single constraint was that quantity q must be non-negative. Adding an oven capacity constraint $q \leq 180$ created a bounded segment with two vertices. Adding a second product and second resource constraint creates a polygon with four vertices. The geometry grows in dimension but the principle remains: the best answer lives at a corner.

Example 8.5.1. (*Extending the baker from Chapter 7*) The baker from Chapter 7 had profit $\pi(q) = 4q - 200$. With the single constraint $0 \leq q \leq 180$, the maximum profit was at $q = 180$: $\pi(180) = \$520$.

Now the baker produces two products: loaves (x) and muffin trays (y), with constraints:

$$3x + 2y \leq 120 \quad (\text{flour}), \quad x + 2y \leq 60 \quad (\text{time}), \quad x \geq 0, \quad y \geq 0.$$

The one-dimensional profitable zone has become the two-dimensional feasible region with vertices $(0, 0)$, $(40, 0)$, $(30, 15)$, $(0, 30)$.

With profit $\pi = 4x + 5y$, the maximum is \$195 at $(30, 15)$. The problem has the same structure as Chapter 7: a convex set, a linear objective, a best answer at a corner.

Exercises 8.5

8.5.1. A food truck from Chapter 7 had $F = \$350$, $v = \$4$, $p = \$9$, and break-even at $q = 70$. Now the truck sells two items: sandwiches (x) and salads (y). Constraints:

$$x + 2y \leq 80 \quad (\text{ingredients}), \quad x + y \leq 60 \quad (\text{prep time}), \quad x \geq 0, \quad y \geq 0.$$

Profit: $\pi = 5x + 8y$.

- (a) Find the vertices of the feasible region.
- (b) Evaluate π at each vertex.
- (c) Identify the optimal production plan.

8.5.2. A tutoring center offers math tutoring (x hours) and writing tutoring (y hours) per week. Constraints:

$$2x + y \leq 40 \quad (\text{tutor availability}), \quad x + 2y \leq 36 \quad (\text{room capacity}), \quad x \geq 0, \quad y \geq 0.$$

Revenue: $R = 30x + 25y$.

- (a) Find all vertices of the feasible region.

- (b) Maximize R .
- (c) At the optimal vertex, how many hours of each type of tutoring are offered?

8.6 The Observer and the Region

The feasible region is determined by the constraints of a situation. It does not change when the observer changes. What changes is the linear expression the observer chooses to evaluate over the region.

Observer	Expression	Action
Profit-focused owner	Maximize $ax + by$	Find maximum vertex
Cost-conscious owner	Minimize $cx + dy$	Find minimum vertex
Revenue analyst	Maximize $px + qy$	Find maximum vertex
Regulator	Bound $ex + fy$	Verify feasibility

Each observer extracts a different answer from the same region. The feasible region is the shared object. The objective is the observer's contribution. This is the same structure seen in Chapters 3 and 4: the data set is shared, but the mean and the median and the z-score each extract different information from it.

Example 8.6.1. A workshop produces chairs (x) and tables (y) subject to:

$$2x + 4y \leq 40, \quad x + y \leq 14, \quad x \geq 0, \quad y \geq 0.$$

The vertices are $(0, 0)$, $(14, 0)$, $(8, 6)$, $(0, 10)$.

The owner asks three different questions.

Question 1: Maximize profit $\pi = 30x + 50y$.

Vertex	$\pi = 30x + 50y$
$(0, 0)$	0
$(14, 0)$	420
$(8, 6)$	540 $\leftarrow$ maximum
$(0, 10)$	500

Optimal: produce 8 chairs and 6 tables. Profit: \$540.

Question 2: Maximize revenue $R = 45x + 80y$.

Vertex	$R = 45x + 80y$
$(0, 0)$	0
$(14, 0)$	630
$(8, 6)$	$840 \leftarrow$ maximum
$(0, 10)$	800

Optimal: same vertex, but the revenue figure differs.

Question 3: Minimize material cost $C = 10x + 20y$.

Vertex	$C = 10x + 20y$
$(0, 0)$	$0 \leftarrow$ minimum
$(14, 0)$	140
$(8, 6)$	200
$(0, 10)$	200

The minimum material cost is zero, at zero production. In practice, the business must cover its fixed costs, so a minimum production level would be imposed by an additional constraint.

The three questions produce three different answers, and two of them land at different vertices. The feasible region did not change.

Exercises 8.6

8.6.1. A garden center sells potted plants (x) and hanging baskets (y). Constraints:

$$x + 2y \leq 60, \quad x + y \leq 40, \quad x \geq 0, \quad y \geq 0.$$

- (a) Find the vertices of the feasible region.
- (b) Maximize profit $\pi = 8x + 15y$.
- (c) Maximize revenue $R = 12x + 20y$.
- (d) Do both objectives lead to the same optimal vertex? Explain in one sentence what this tells the garden center owner.

8.6.2. Using the workshop constraints from the main example in this section (vertices $(0, 0)$, $(14, 0)$, $(8, 6)$, $(0, 10)$):

- (a) Find the expression $f = ax + by$ whose maximum occurs at $(14, 0)$ rather than $(8, 6)$. *Hint: try $f = 50x + 10y$ and verify.*
- (b) In one sentence, describe the type of situation (in terms of product values) that would make vertex $(14, 0)$ optimal.

Chapter 8 Review

6. Determine whether each set is convex. Justify each answer in one sentence.

- (a) The set of all points satisfying $x \geq 0$, $y \geq 0$, $x + y \leq 10$.
- (b) The set of points inside two non-overlapping circles.
- (c) A solid parallelogram.
- (d) The set of all points satisfying $x^2 + y^2 \leq 9$ (a solid disk).

6. Sketch the feasible region for each system. Label all vertices.

- (a) $x + y \leq 12$, $2x + y \leq 18$, $x \geq 0$, $y \geq 0$
- (b) $x + 2y \leq 16$, $3x + 2y \leq 24$, $x \geq 0$, $y \geq 0$

6. A coffee shop sells lattes (x) and cappuccinos (y) subject to:

$$2x + 3y \leq 90 \quad (\text{milk}), \quad x + y \leq 40 \quad (\text{cups available}), \quad x \geq 0, \quad y \geq 0.$$

- (a) Find all vertices of the feasible region.
- (b) Maximize profit $\pi = 3x + 4y$.
- (c) Maximize revenue $R = 5x + 6y$.
- (d) Do both objectives lead to the same optimal vertex? Explain.

6. A nutritionist plans daily servings of two food sources, A and B. Each serving of A provides 4 grams of protein and 2 grams of fiber. Each serving of B provides 2 grams of protein and 5 grams of fiber. Constraints require at least 20 grams of protein ($4x + 2y \geq 20$) and at least 15 grams of fiber ($2x + 5y \geq 15$), with $x \geq 0$ and $y \geq 0$.

- (a) Find the vertices of the feasible region. *Note: the inequalities point the other direction, so the feasible region is unbounded. Find the vertices at the intersections of the boundary lines and with the axes.*
- (b) Minimize total cost $C = 3x + 2y$.

6. The feasible region of a system has vertices $(0, 0)$, $(10, 0)$, $(7, 4)$, and $(0, 8)$.

- (a) For what values of a does $f = ax + 3y$ achieve its maximum at $(0, 8)$? *Hint: evaluate at $(7, 4)$ and $(0, 8)$ and set up an inequality.*
 - (b) Evaluate $f = 2x + 5y$ at all four vertices and identify where the maximum occurs.
6. Explain in three complete sentences the connection between the break-even analysis of Chapter 7 and the feasible region analysis of Chapter 8. Include a description of how the dimensionality of the problem changes and what stays the same.